

# SKILL CORTEX

## Product Requirements Document (PRD)

**Product Version:** V1.0

**Document Version:** 1.0

**Product Type:** Webinar & Course Booking Platform

**Status:** Product Definition

**Primary Platforms:** Web Application

**Primary Users:** Learners / Students

**Administrative Users:** Skill Cortex Admin

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## 1. Product Overview

**Skill Cortex** is a web-based webinar and course booking platform that enables users to discover educational programs, select their department, choose a webinar and available time slot, complete online payment through Razorpay, and receive automated booking confirmations and reminders.

The product consists of two major experiences:

### User Application

Allows users to:

- Discover webinars
- Register/login
- Select department
- Select courses
- Select slots
- Book webinars
- Make payments
- View bookings
- Receive notifications
- Receive reminders

### Admin Application

Allows administrators to:

- Manage users
  - Manage departments
  - Manage webinars
  - Manage slots
  - Monitor bookings
  - Monitor payments
  - Monitor notifications
-

## 2. Product Vision

**Make educational webinar registration simple, secure, automated and accessible from discovery to attendance.**

The core product journey is:

**Discover → Register → Select → Book → Pay → Confirm → Remember → Attend**

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## 3. Product Goals

### Primary Goals

#### Goal 1 — Easy Discovery

Users should be able to understand Skill Cortex and discover available courses from the landing page.

#### Goal 2 — Simple Booking

A user should be able to book a webinar through a straightforward workflow.

#### Goal 3 — Secure Payment

Users should be able to pay online using Razorpay.

#### Goal 4 — Reliable Confirmation

A booking should be confirmed only after successful payment verification.

#### Goal 5 — Automated Reminders

Users should receive reminders automatically before their webinar.

#### Goal 6 — Efficient Administration

Administrators should be able to manage the complete webinar lifecycle from one dashboard.

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## 4. Product Success Metrics

The following metrics can be tracked after launch:

| Metric                       | Objective                            |
|------------------------------|--------------------------------------|
| Registration completion rate | Increase successful account creation |

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| Metric                    | Objective                          |
|---------------------------|------------------------------------|
| Booking completion rate   | Increase successful bookings       |
| Payment success rate      | Monitor payment performance        |
| Slot utilization          | Measure webinar seat usage         |
| Reminder delivery rate    | Ensure reminders reach users       |
| Booking cancellation rate | Monitor booking reliability        |
| Webinar attendance rate   | Measure effectiveness of reminders |
| Admin processing time     | Reduce manual administration       |

## 5. Target Users

### Primary Persona — Learner

A student or professional looking for technical training.

#### Needs

- Discover relevant courses
- Know course price
- Choose convenient timing
- Make payment easily
- Receive confirmation
- Remember the session

#### Pain Points

- Manual registration
- Unclear slot availability
- Payment uncertainty
- Missing webinar timings
- Lack of booking history

### Secondary Persona — Administrator

The person responsible for managing Skill Cortex.

#### Needs

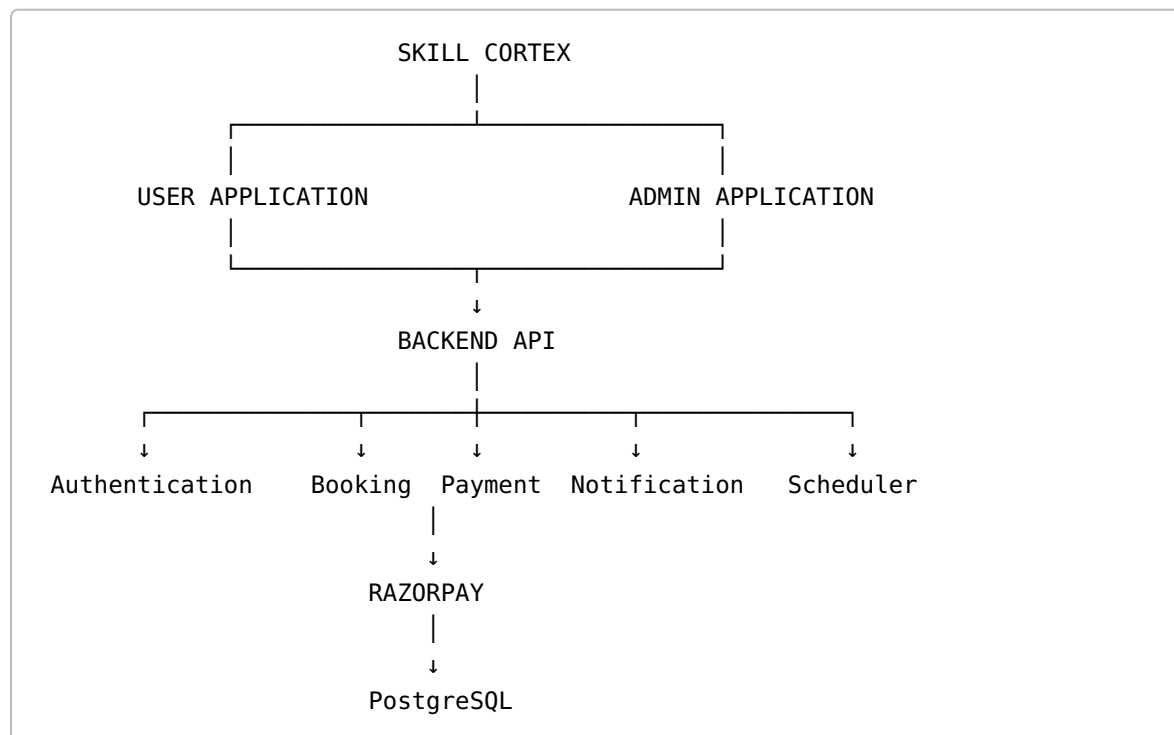
- Manage courses
- Manage slots
- Monitor registrations
- Track payments

- Communicate with users

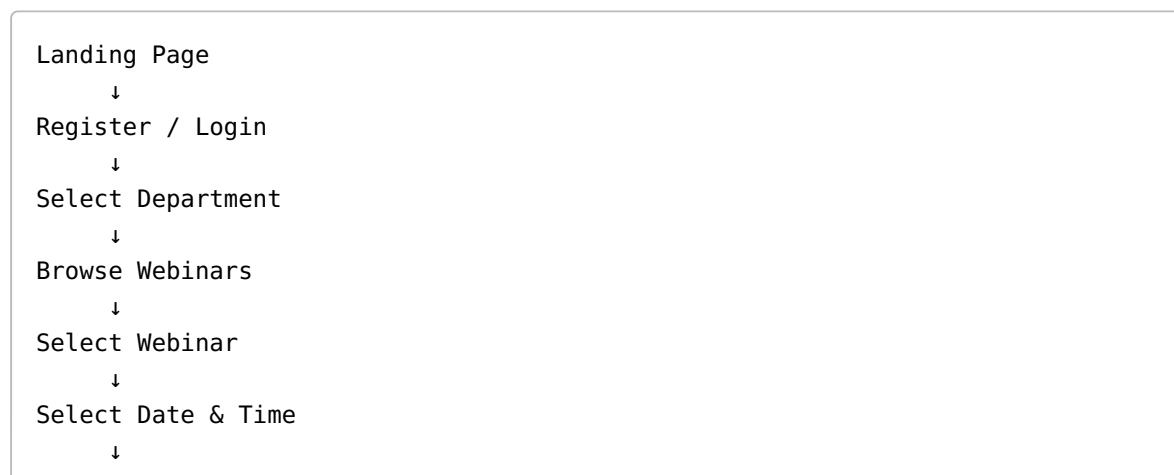
### Pain Points

- Manual payment tracking
- Manual reminders
- Difficulty monitoring seats
- Scattered user information

## 6. Product Architecture



## 7. User Experience Flow



```
Review Booking
  ↓
Razorpay Checkout
  ↓
Payment Verification
  ↓
Booking Confirmation
  ↓
Email / SMS
  ↓
Automated Reminders
  ↓
Webinar
```

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## 8. Feature Requirements

### 8.1 Landing Page

#### Objective

Introduce Skill Cortex and allow visitors to discover available training programs.

#### Requirements

The landing page must contain:

- Skill Cortex logo/name
- Navigation
- Hero section
- Platform introduction
- Featured webinars
- Department information
- Benefits
- Call-to-action
- Login button
- Register button
- Contact/footer section

#### Primary CTA

##### Explore Webinars

Secondary CTA:

##### Register Now

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## 8.2 User Registration

### Objective

Allow new users to create an account.

### Required Fields

- Full name
- Email
- Mobile number
- Password
- Department

### Validation

The system must validate:

- Required fields
- Email format
- Phone format
- Password requirements
- Duplicate email

### Success

After registration:

### Account Created Successfully

The user can proceed to login.

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## 8.3 User Login

Users shall enter:

- Email
- Password

After successful authentication:

### User → User Dashboard

Invalid credentials should display a meaningful error message.

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## 8.4 User Dashboard

The dashboard shall provide:

### Welcome Section

Example:

Welcome back, Dhanushya!

### Upcoming Webinar

Display:

- Webinar
- Date
- Time
- Booking status

### Quick Actions

- Browse Webinars
- My Bookings
- Profile

### Booking Summary

Display:

- Upcoming bookings
- Completed bookings
- Payment status

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## 8.5 Department Selection

Users shall be able to select a department.

Example:

CSE  
AI / ML  
Data Science

The system shall use department mapping to display relevant webinars.

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## 8.6 Webinar Listing

The webinar page shall display available courses.

Each card should contain:

- Course title
- Short description
- Department
- Price
- Duration
- Next available slot
- Available seats
- View Details button

Example:

Python Programming

Learn Python fundamentals and practical programming.

₹999

Next Slot:

20 September – 10:00 AM

[View Details]

---

## 8.7 Webinar Details

The detail page should provide:

- Webinar title
- Full description
- Instructor
- Duration
- Department
- Price
- Available slots
- Available seats
- Booking CTA

Primary CTA:

**Book Now**

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## 8.8 Slot Selection

The user shall see available slots.

Example:

| Date   | Time     | Seats | Action   |
|--------|----------|-------|----------|
| 20 Sep | 10:00 AM | 15    | Select   |
| 21 Sep | 6:00 PM  | 8     | Select   |
| 22 Sep | 10:00 AM | Full  | Disabled |

Unavailable slots must not be selectable.

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## 8.9 Booking Review

Before payment, display:

### Booking Summary

Webinar: Python Programming

Department: CSE

Date: 20 September 2026

Time: 10:00 AM

Price: ₹999

Total: ₹999

[Proceed to Payment]

The user should confirm the information before proceeding.

---

## 8.10 Payment

### Payment Provider

Razorpay

Flow

Create Booking



```
Create Razorpay Order
  ↓
Open Checkout
  ↓
Payment
  ↓
Payment Response
  ↓
Backend Verification
  ↓
Update Payment
  ↓
Confirm Booking
```

## 8.11 Payment Verification

The backend must verify payment information.

### Successful Payment

```
Payment = PAID
Booking = CONFIRMED
```

### Failed Payment

```
Payment = FAILED
Booking ≠ CONFIRMED
```

The system must not confirm a booking solely because the frontend reports success.

## 8.12 Booking Confirmation

After successful payment verification, the user shall see:

### Booking Confirmed

Information:

- Booking ID
- Webinar
- Date
- Time
- Amount paid

- Payment reference

The confirmation page should provide:

**View My Booking**

---

## 8.13 My Bookings

Users shall be able to view:

### Upcoming

- Webinar
- Date
- Time
- Booking status
- Payment status

### Previous

- Webinar
  - Date
  - Attendance/session status, if supported
- 

## 8.14 Email Notification

After successful payment:

The system sends a confirmation email.

### Email should include:

- User name
  - Webinar
  - Date
  - Time
  - Amount
  - Booking ID
  - Payment reference
- 

## 8.15 SMS Notification

The system should send an SMS confirmation after verified payment.

The SMS should be short and contain:

- Webinar
- Date/time
- Booking confirmation
- Amount or reference where appropriate

---

## 8.16 Admin Notification

Admin receives notification after successful payment.

Example:

```
New Payment Received
```

```
User: John Doe
```

```
Course: Python Programming
```

```
Slot: 20 Sep, 10:00 AM
```

```
Amount: ₹999
```

```
Status: PAID
```

```
Booking ID: SC-10025
```

---

## 8.17 Reminder System

The scheduler will periodically check upcoming confirmed bookings.

Example:

```
Booking
```

```
↓
```

```
Check webinar date/time
```

```
↓
```

```
Calculate reminder schedule
```

```
↓
```

```
Reminder due?
```

```
↓
```

```
Yes
```

```
↓
```

```
Send Email/SMS
```

```
↓
```

```
Record notification
```

## 8.18 Reminder Configuration

Initial reminder schedule:

- 3 days before
- 2 days before
- 1 day before
- Day of webinar

The schedule should be configurable by the system administrator in future iterations.

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## 8.19 Admin Dashboard

The admin dashboard should display:

Total Users  
Total Webinars  
Upcoming Sessions  
Total Bookings  
Successful Payments  
Pending Payments  
Failed Payments  
Revenue

Where revenue reporting is enabled.

---

## 8.20 User Management

Admin can:

- View users
  - Search users
  - Filter by department
  - View user details
  - View user bookings
  - View payment history
- 

## 8.21 Webinar Management

Admin can:

- Create webinar
- Edit webinar

- Delete/deactivate webinar
  - Set price
  - Set duration
  - Add description
  - Assign department
  - Manage status
- 

## 8.22 Slot Management

Admin can:

- Create slots
  - Edit slots
  - Set date
  - Set start/end time
  - Set capacity
  - View availability
  - Activate/deactivate slots
- 

## 8.23 Booking Management

Admin can:

- View bookings
  - Search bookings
  - Filter bookings
  - View booking details
  - View payment status
- 

## 8.24 Payment Management

Admin can view:

- Payment ID
  - Booking ID
  - User
  - Webinar
  - Amount
  - Razorpay order ID
  - Razorpay payment ID
  - Payment status
  - Payment date
-

## 8.25 Notification Management

The system shall maintain notification records.

Each record should contain:

- Recipient
- Booking
- Notification type
- Channel
- Status
- Timestamp
- Failure reason, if applicable

---

## 9. Functional Requirements

| ID     | Feature                      | Priority |
|--------|------------------------------|----------|
| PR-001 | Landing page                 | P0       |
| PR-002 | User registration            | P0       |
| PR-003 | User login                   | P0       |
| PR-004 | Admin authentication         | P0       |
| PR-005 | User dashboard               | P0       |
| PR-006 | Department selection         | P0       |
| PR-007 | Webinar listing              | P0       |
| PR-008 | Webinar details              | P0       |
| PR-009 | Slot selection               | P0       |
| PR-010 | Booking                      | P0       |
| PR-011 | Razorpay payment             | P0       |
| PR-012 | Backend payment verification | P0       |
| PR-013 | Booking confirmation         | P0       |
| PR-014 | Email notification           | P0       |
| PR-015 | SMS notification             | P1       |
| PR-016 | Automated reminders          | P0       |
| PR-017 | My bookings                  | P0       |
| PR-018 | Admin dashboard              | P0       |
| PR-019 | User management              | P1       |

| ID     | Feature              | Priority |
|--------|----------------------|----------|
| PR-020 | Webinar management   | P0       |
| PR-021 | Slot management      | P0       |
| PR-022 | Payment management   | P0       |
| PR-023 | Notification history | P1       |

**P0:** Required for V1

**P1:** Important but can follow the initial MVP

## 10. Role-Based Access

| Feature           | Guest | User | Admin |
|-------------------|-------|------|-------|
| Landing page      | ✓     | ✓    | ✓     |
| Browse webinars   | ✓     | ✓    | ✓     |
| Register          | ✓     | —    | —     |
| Login             | ✓     | ✓    | ✓     |
| Book webinar      | —     | ✓    | —     |
| Make payment      | —     | ✓    | —     |
| View own bookings | —     | ✓    | —     |
| Manage webinars   | —     | —    | ✓     |
| Manage slots      | —     | —    | ✓     |
| View all bookings | —     | —    | ✓     |
| View payments     | —     | —    | ✓     |
| Manage users      | —     | —    | ✓     |

## 11. Business Rules

### Rule 1

Only authenticated users can book webinars.

### Rule 2

Only active webinars can be booked.

**Rule 3**

Only available slots can be selected.

**Rule 4**

The system must prevent overbooking.

**Rule 5**

Payment must be verified on the backend.

**Rule 6**

Only successfully verified payments can confirm bookings.

**Rule 7**

Notifications are triggered after successful payment verification.

**Rule 8**

Reminders are sent only for confirmed bookings.

**Rule 9**

Duplicate reminders must not be sent.

**Rule 10**

Admin-only functionality must be protected by authorization.

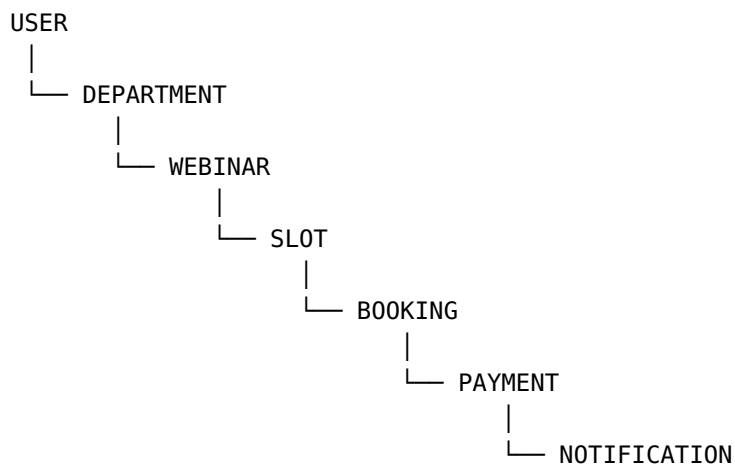
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## 12. Database Requirements

The initial database shall contain:

```
users
departments
webinars
webinar_departments
slots
bookings
payments
notifications
```

## Relationship



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## 13. API Requirements

The backend should expose REST APIs.

### Authentication

```
POST /auth/register
POST /auth/login
POST /auth/logout
POST /auth/forgot-password
```

### Users

```
GET /users/me
PUT /users/me
```

### Departments

```
GET /departments
GET /departments/{id}
```

### Webinars

```
GET /webinars
GET /webinars/{id}
POST /admin/webinars
```

```
PUT /admin/webinars/{id}
DELETE /admin/webinars/{id}
```

## Slots

```
GET /webinars/{id}/slots
POST /admin/slots
PUT /admin/slots/{id}
DELETE /admin/slots/{id}
```

## Bookings

```
POST /bookings
GET /bookings
GET /bookings/{id}
```

## Payments

```
POST /payments/create-order
POST /payments/verify
```

## Notifications

```
GET /notifications
```

Admin notification APIs can be added separately.

---

# 14. Payment Requirements

The payment implementation must:

- Generate unique Razorpay orders.
  - Associate orders with bookings.
  - Store Razorpay identifiers.
  - Verify payment signatures/server-side responses.
  - Handle failed payments.
  - Prevent duplicate confirmation.
  - Maintain payment audit information.
- 

# 15. Notification Requirements

The notification system should support:

### **Immediate Notifications**

Triggered after payment verification.

### **Scheduled Notifications**

Triggered according to the reminder schedule.

### **Notification Status**

PENDING  
SENT  
DELIVERED  
FAILED

Where the provider supports delivery status.

---

## **16. Error Handling**

The product should provide meaningful messages.

### **Examples**

#### **Invalid Login**

Email or password is incorrect.

#### **Full Slot**

This slot is full. Please choose another available slot.

#### **Payment Failed**

Payment could not be completed. Please try again.

#### **Payment Verification Pending**

Your payment is being verified. Please check your booking status shortly.

#### **Webinar Unavailable**

This webinar is currently unavailable.

---

## 17. Security Requirements

The application shall:

- Hash passwords securely.
  - Protect authenticated routes.
  - Implement role-based authorization.
  - Validate API inputs.
  - Protect secrets using environment variables.
  - Use HTTPS in production.
  - Protect payment verification endpoints.
  - Prevent unauthorized access to user data.
  - Avoid exposing sensitive payment credentials.
- 

## 18. Performance Requirements

The system should:

- Provide responsive UI interactions.
  - Handle multiple simultaneous users.
  - Handle concurrent slot booking safely.
  - Process payment responses reliably.
  - Execute scheduled reminders efficiently.
- 

## 19. Reliability Requirements

The system must preserve transactional data.

Payment success should not be lost if a notification service fails.

For example:

```
Payment Verified ✓  
  ↓  
Booking Confirmed ✓  
  ↓  
Email Failed ✗  
  ↓  
Notification marked FAILED
```

The booking must remain confirmed.

---

## 20. Admin Operational Requirements

The admin should be able to perform the following without developer assistance:

```
Create Department
      ↓
Create Webinar
      ↓
Set Price
      ↓
Create Slots
      ↓
Monitor Bookings
      ↓
Monitor Payments
      ↓
Monitor Notifications
```

---

## 21. UI/UX Requirements

The UI should be:

- Clean
- Modern
- Responsive
- Mobile-friendly
- Accessible
- Easy to navigate

The main booking journey should require minimal unnecessary steps.

Important actions should have clear CTAs:

- Explore Webinars
  - Register
  - Login
  - View Details
  - Book Now
  - Select Slot
  - Proceed to Payment
  - View Booking
-

## 22. Responsive Requirements

The product should support:

- Desktop
- Laptop
- Tablet
- Mobile

The booking process should remain usable on smaller screens.

---

## 23. MVP Definition

The minimum viable product consists of:

- ✓ Landing Page
  - ✓ Registration/Login
  - ✓ Department Selection
  - ✓ Webinar Catalog
  - ✓ Webinar Details
  - ✓ Slot Selection
  - ✓ Booking
  - ✓ Razorpay Payment
  - ✓ Payment Verification
  - ✓ Booking Confirmation
  - ✓ Email Notification
  - ✓ SMS Notification
  - ✓ Automated Reminders
  - ✓ User Dashboard
  - ✓ Admin Dashboard
  - ✓ Webinar Management
  - ✓ Slot Management
  - ✓ Booking Management
  - ✓ Payment Monitoring
- 

## 24. V1 Exclusions

The following are intentionally excluded:

- AI skill analysis
- Resume parsing
- AI course recommendations
- AI career recommendations
- AI interview preparation
- Job matching

- AI agents
- Advanced learning management
- Advanced analytics

These features belong to future releases.

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## 25. Future Roadmap

### V2 — Skill Intelligence

Introduce:

- User skill profiles
- Skill assessment
- Skill-gap analysis
- Course recommendations

### V3 — Career Intelligence

Introduce:

- Resume analysis
- Job matching
- Career roadmap
- Interview preparation

### V4 — Agentic Skill Cortex

Introduce:

- AI learning assistant
  - Personalized learning agent
  - Automated learning plans
  - Skill progression monitoring
  - Intelligent recommendations
- 

## 26. Product Acceptance Criteria

Skill Cortex V1 will be considered ready when the following complete journey works:

**User**

**Landing Page**

↓

**Register**



**Login**



**Select Department**



**Select Webinar**



**Select Slot**



**Review Booking**



**Pay with Razorpay**



**Payment Verified**



**Booking Confirmed**



**Confirmation Email/SMS**



**Automated Reminder**



**Attend Webinar**

## Admin

At the same time, the administrator must be able to:

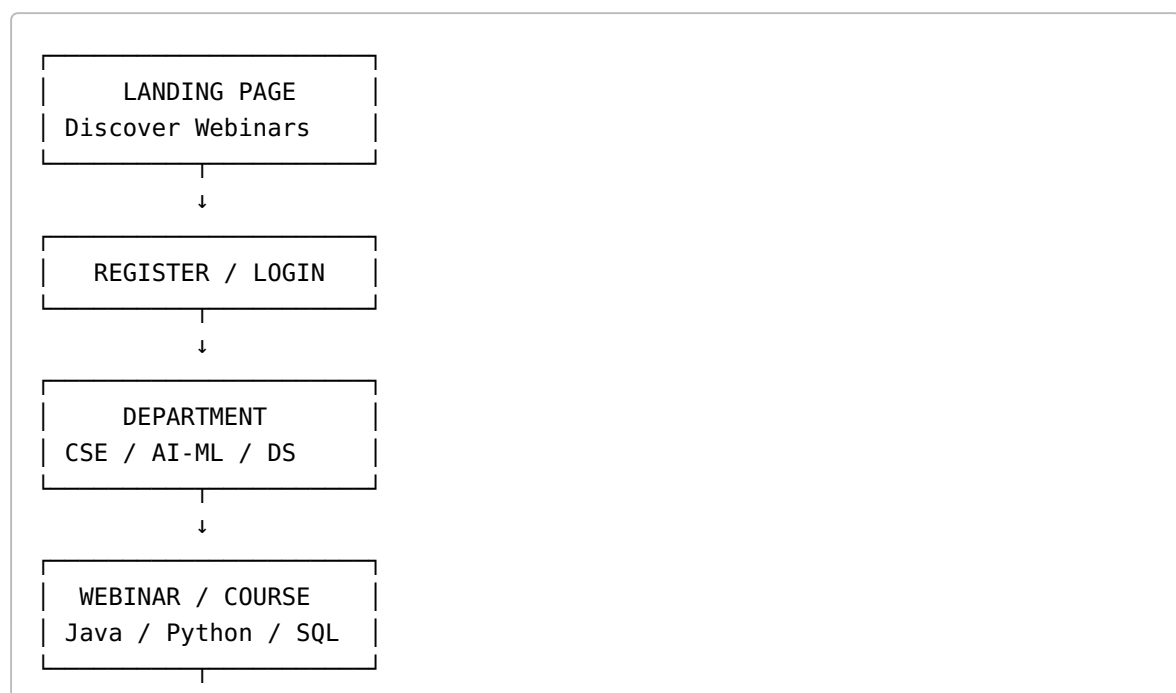
- See the new user.
  - See the booking.
  - See the verified payment.
  - See the amount paid.
  - Receive the payment notification.
  - Manage the webinar and slot.
- 

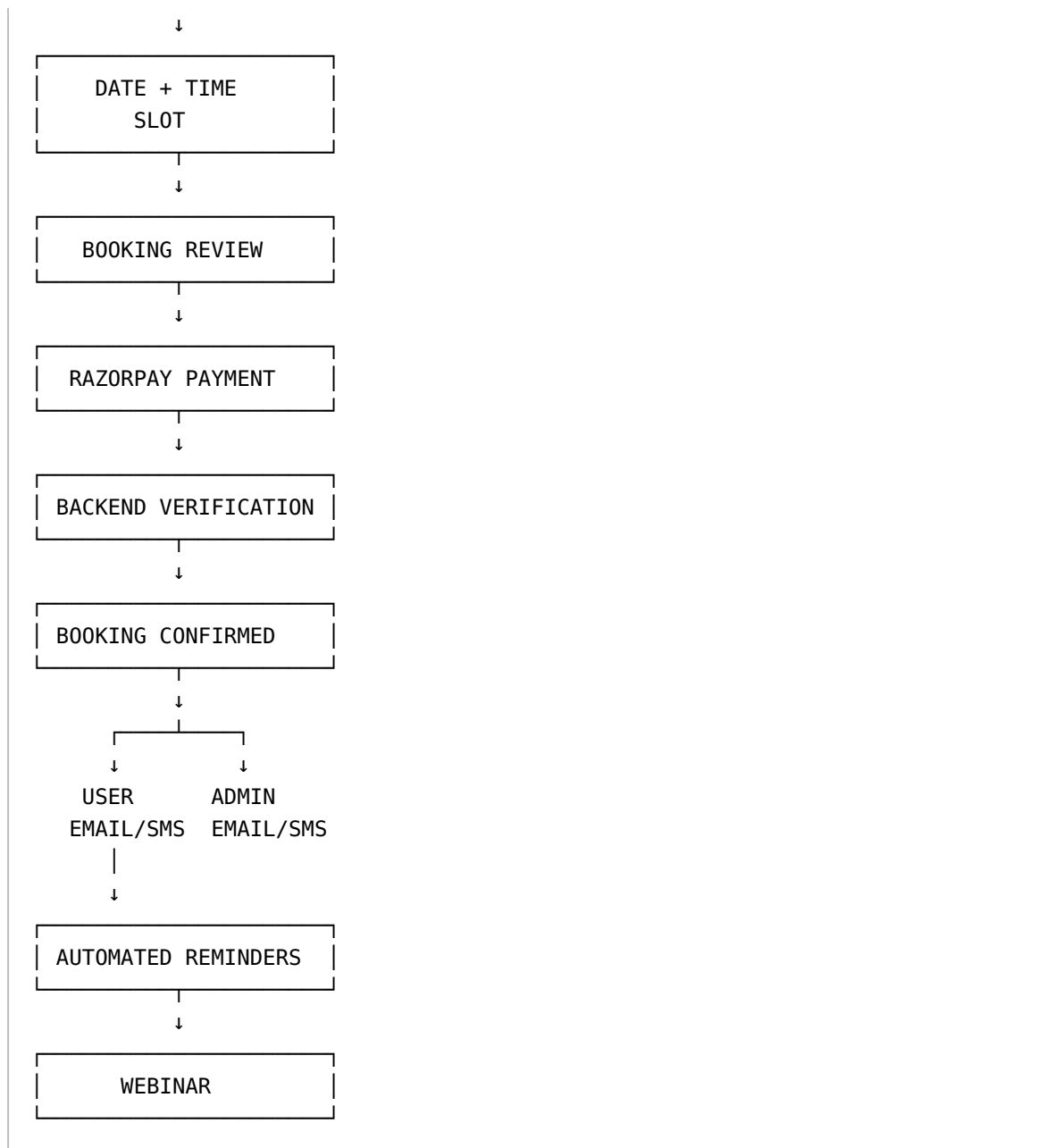
## 27. Definition of Done

A feature is considered complete only when:

1. UI is implemented.
  2. Backend API is implemented.
  3. Database integration is complete.
  4. Authentication/authorization is applied where required.
  5. Validation is implemented.
  6. Error handling is implemented.
  7. Relevant notification behavior is implemented.
  8. Tests are completed.
  9. The feature works on desktop and mobile.
  10. The feature is integrated into the complete V1 workflow.
- 

## 28. Final Product Flow





## 29. Product Summary

**Skill Cortex V1** is a complete webinar and course booking product designed around a simple but reliable user journey.

The product combines:

**Course Discovery + Department Selection + Slot Booking + Online Payment + Payment Verification + Notifications + Automated Reminders + Administration**

The V1 architecture deliberately establishes the foundation for future versions in which Skill Cortex can evolve from a booking platform into an intelligent **skill development and career platform**.